

TaSain Thomas

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EDUCATION

Wayne State University B.S. Computer Science | May 2026 | GPA: 3.4/4.0

Trustworthy AI Coursework: LoRA/QLoRA, RLHF/DPO/GRPO, Red Teaming, Machine Unlearning, Agentic AI Safety, Grounded VLMs, Chain-of-Thought Reasoning, HarmBench Benchmarking

CORE COMPETENCIES

ML Validation • CI/CD • Red Teaming • Model Monitoring • LoRA/VLM Fine-Tuning • RAG • Simulation-Based Evaluation • Uncertainty Quantification • Kubernetes • Docker • Python • PyTorch • C++ • Adversarial Robustness

PROFESSIONAL EXPERIENCE

Industry 4.0 / Automation Intern (Rotational) | *Magna EV Structures* | Sep 2023 • Sep 2024

- Delivered 4 automation projects on schedule by integrating real-time production data pipelines via custom API development
- Improved throughput 33% by implementing Vuforia Model Targets for vision-guided assembly verification
- Built C++/SQLite3 worker-to-station matching engine with evaluateFit() scoring algorithm, reducing assignment time 40%
- Reduced debug cycle time 25% by developing custom C# validation logic that automated error handling and root-cause identification

Registered Behavior Technician | *BlueMind Therapy LLC* | Sep 2024 • Present

ENGINEERING PROJECTS

BMI Estimator with Reasoning: Conformal Prediction + GPT-as-Judge | Apr 2026

- Trained a Squeeze-and-Excitation DenseNet (SE-DenseNet121) in PyTorch on Celeb-FBI (7,208 images), achieving 2.81 BMI-point MAE (11.56% MAPE)
- Added split conformal prediction for calibrated uncertainty: 90% intervals with 90.4% verified coverage on held-out data
- Layered a GPT-4.1-Nano LLM-as-judge reasoning module that explains each estimate and flags edge cases

MLOps Infrastructure & AI Agent Engineering | Mar 2026 • Present

- Architected auto-scaling Kubernetes inference cluster with GPU passthrough serving LLMs at 99.9% uptime, with model monitoring and real-time incident response
- Embedded red-team security testing (Kali Linux) into CI/CD, patching prompt injection vulnerabilities pre-production

SaaS Booking & Payment Platform | github.com/T-Thomas1/on-wheels_detailing-crm

- Engineered full-stack CRM (Python/JavaScript) with Stripe payments and Telegram Bot API; deployed on DigitalOcean, Nginx, systemd, Cloudflare CI/CD at 24/7 uptime

Real-Time Mobile Collaboration Platform | Capstone, May 2026

- Developed cross-platform (Flutter/Dart) collaborative workspace integrating ZegoCloud real-time video and RevenueCat subscription monetization; shipped to production for client deployment

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Java, Dart

ML / AI: PyTorch, TensorFlow, Hugging Face, LangChain, LoRA/QLoRA, RAG, Vector Databases, Transformer Architectures

MLOps: Kubernetes, Docker, CI/CD, Model Serving & Monitoring, Helm, Terraform, GPU Passthrough

Cloud / Infra: AWS/Azure, DigitalOcean, Nginx, Cloudflare, Proxmox, systemd

Security: Red Teaming, Prompt Injection Defense, Zero Trust, Adversarial Robustness, Kali Linux